Data and Information Quality - Project Report

Project ID: 47

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1 Setup Choices

Libraries Used: To handle the data and create visualizations, we utilized standard Python libraries including `numpy`, `pandas`, `seaborn`, and `matplotlib`.
For imputing missing values, the `KNNImputer` from `scikit-learn` was employed.
Outlier detection was conducted using `DBSCAN`, also provided by `scikit-learn`.
Near-duplicate records were identified using the `recordlinkage` library, while fuzzy string matching was performed using the `rapidfuzz` library.
No additional libraries were used in the project.

2 Pipeline Implementation

2.1 Data Profiling and Quality Assessment

We have two datasets to clean: `artists.csv` and `artworks.csv`. The shapes of the datasets are the following:

- Artists Dataset shape: (15091, 6)
- Artworks Dataset shape: (130262, 21)

They share two columns: `Name` and `Artist ID`, which we used to merge the dataframes. We observed that there are more unique IDs in `artists.csv` than in `artworks.csv`.
In both datasets, there are no exact duplicates.

Completeness of the datasets was computed as follows:

- Completeness of `artists.csv`: 78.0%
- Completeness of `artworks.csv`: 69.3%

2.2 Data Cleaning

The first step involved merging the datasets using `Artist ID` as the common key. To confirm the success of the merge, we checked the consistency of the `Name` columns across both datasets. The resulting dataframe consists of 25 columns.

2.2.1 Data Transformation/Standardization

- **Replacing Null-Like Values:** Null-like values such as `"unknown"`, `"N/A"`, and `"-"` were replaced with `NaN` for consistent handling of missing data.
- **Standardizing Categorical Values:** The `Gender` column had inconsistent values such as `"male"` and `"Male"`. These were standardized to `"Male"`.
- **Converting Date Columns:** The `Birth Year` and `Death Year` columns were converted from `float` to `int`, ensuring no `Birth Year` exceeded its corresponding `Death Year`. Invalid entries were replaced with `NaN`.

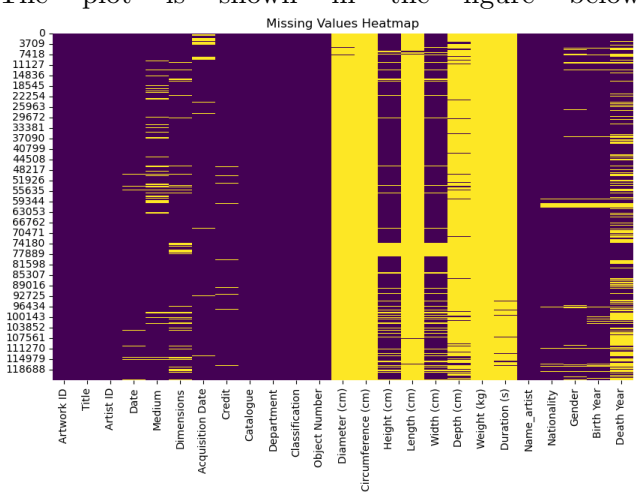
- **Nationality Correction:** Entries such as `"Nationality Unknown"` and `"Nan"` were replaced with `NaN`. Remaining values were stripped of extra spaces and title-cased (e.g., `"american"` to `"American"`).

- **Standardizing Date Formats:** The `Date` and `Acquisition Date` columns were reformatted to `YYYY-MM-DD`. Missing dates were imputed with the median.

- **Detecting and Correcting Typos:** Using `rapidfuzz`, we explored fuzzy matching with a similarity threshold of 95% to detect and correct typos in the `Medium` column. While effective for minor variations, it struggled with numeric differences (e.g., `"Oil on canvas 1"` vs. `"Oil on canvas 2"`). Thus, it was explored but not implemented.

2.2.2 Missing Values

To identify the extent of missing data, we visualized the missing values across all columns. The plot is shown in the figure below:



From the visualization, we observed that certain columns, such as `Diameter (cm)`, `Length (cm)`, `Depth (cm)`, `Weight (kg)`, `Duration (s)`, and `Circumference (cm)`, had almost no data. Given the negligible completeness of these columns, we decided to drop them.

For the `Death Year` column, missing values were likely indicative of artists who are still alive. To maintain consistency and allow for easy interpretation, we replaced these missing values with `9999`.

For the `Title`, `Medium`, and `Credit` columns, there were no feasible methods to predict or derive the missing values. Thus, we opted to replace them with placeholder values, ensuring the dataset remained analyzable without introducing bias.

The `Gender` and `Nationality` columns were categorical, and we addressed their missing values by filling them with the most common values, which were `Male` for `Gender` and `American` for `Nationality`.

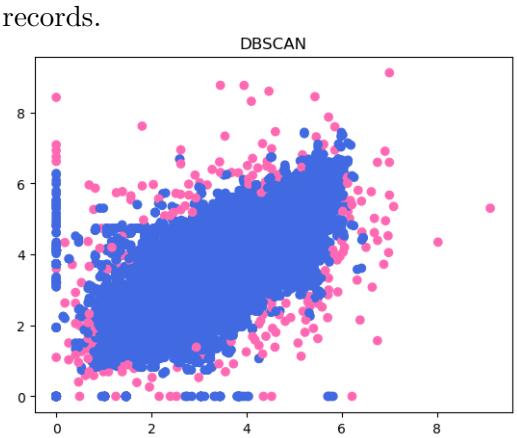
For numerical columns such as `Height`, `Width`, and `Birth Year`, we employed the `k-nearest neighbors` (KNN) imputation technique. Additionally, we ensured logical consistency by verifying that no computed `Birth Year` exceeded the corresponding `Death Year`.

The `Dimensions` column was partially derived from `Height` and `Width`. To complete the missing values, we used the pattern `(Height * Width)`, which consistently appeared in existing records.

Finally, for the `Acquisition Date` and `Date` columns, missing values were replaced with the median.

2.2.3 Outliers

Outliers were detected using the `DBSCAN` algorithm, focusing on the dimensions of the artwork (height and width). The data was transformed using `log1` prior to applying `DBSCAN` to stabilize variance and improve detection accuracy. Outliers were not removed, as the dataset is intended for storage rather than prediction, aligning with the goal of preserving all records.



3 Data Deduplication

- **Exact Matching:** We first searched for duplicates using exact matching but found no duplicate records in the dataset.

- **Record Linkage:** We explored the use of the `recordlinkage` library to identify near-duplicate records based on columns like `Name_artist` and `Title`.
  - The method used a sorted neighborhood indexing approach and Jaro-Winkler similarity for string comparisons.
  - However, no meaningful duplicates were detected, and this approach was not implemented in the dataset.

4 Additional Data Quality Assessment

After cleaning the data, we verified that the desired quality level was achieved through the following checks:

- **Missing Values:** The majority of columns had no missing values, except for `Name_artist`, which contained 15 missing entries. This represents a negligible percentage (0.01%) of the dataset and does not significantly impact analysis.
- **Duplicate Records:** No duplicate rows were found in the dataset, ensuring that all entries are unique.
- **Consistent Formatting:**
  - Column types were confirmed to match expected formats (e.g., `Date` and `Acquisition Date` are in `datetime` format, and numeric columns such as `Height (cm)` and `Width (cm)` are correctly typed as `float`).
  - All categorical columns were standardized and free of formatting inconsistencies.
- **Dataset Summary:** The final dataset contains 122,373 records with 19 columns. The cleaned data meets the desired quality standards for completeness, consistency, and uniqueness.

References

[1] Cinzia Cappiello, *Data and Information Quality*, Lecture slides, Politecnico di Milano, WeBeep platform.